Reference: AOEC's Research Paper submission for DCAL 2026

Hypothesis: Issue information reporting for Packaged RS&A related does have an impact on V2R effectiveness, responsiveness and planned performance

Description:

As there are 3 main categories of risk and live categories of fire ratings, the insight to fire mitigation intelligence is to develop packages that can be deployed or incorporated into or instantiated at locations.

RS&A-RCS packaging for V2R effectiveness for fire incidences that are known/earlier-known or aerially-surveyed/reported is a solution that is cost-effective, that has rapid fire detection, has sense & respond capabilities of early warning, with analytics for ease in implementation and integration with other fire safety systems.

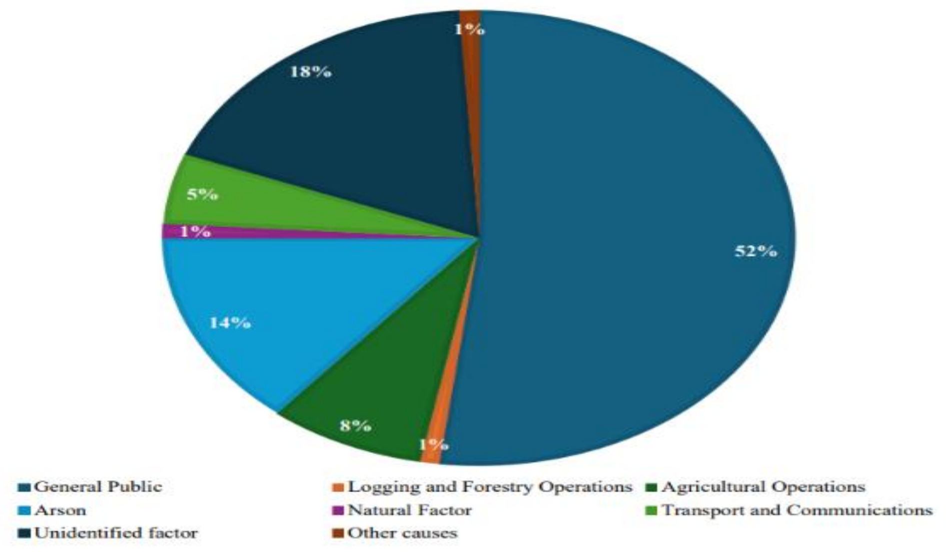
The 3 main categories of risk and the type of FP&S needed during a fire incidence

Type A: Fire prevention and extinguisher products need to put out fires involving Paper, wood, cloth and plastic more generally

Type 8: Fire prevention and extinguisher products need to put out fires involving Petrol, oil paints, spirit, chemicals more generally

Type C: Fire prevention and extinguisher products need to put out fires involving Cooking & welding gas, electrical fires, faulty live-electrical-equipment more generally

Connecting all this further: RS&A-RCS packaging often needs to address a past research paper's indication of percentages of fires and causes



Bulletin Board incorporation of fire hazards and risks, or fire mitigation intelligence can help RS&A-RCS packaging for fire incidences, or their fire detection and prevention/mitigation

In fire safety, time is everything. The faster a fire spreads, the less time people have to escape, and the harder it becomes to control the situation. That's where flame spread ratings come into play.

Flame spread rating indicate how quickly flames will travel along the surface of a given material.

It's one of the key fire rating classifications used in both residential and commercial construction, helping code officials, designers, and inspectors assess the relative fire hazard of different building materials.

Flame spread ratings are based on the Flame Spread Index (FSI), a numerical value derived from a standardized test that ranks how far and fast flames can move across a material's surface. Materials with lower FSI values are less likely to play a significant part in spreading fires, which typically makes them a safer option.

Depending upon its specific relevance, one of the most commonly used fire spread test method is ASTM E84, also known as the Steiner Tunnel Test.

Recognized by code bodies like the International Building Code (IBC), ASTM E84 is often required for code approval in construction/dashboarding/bulletin board systems.

ASTM E84 has two important measurements that will come from the test results: Flame Spread Index (FSI) and Smoke Developed Index (SDI).

FSI quantifies the rate and extent of flame spread, while SDI measures the amount of smoke produced. These values then help to classify a material's surface burning characteristics, which determine where it can be used within a building based on the fire rating class.

There are three classifications, based on specified ranges of flame spread and smoke developed indices, that typically come out of the ASTM E84 test.

Class A identifies the best performing materials, falling within the range of 0 to 25 FSI. Class B is mid-range, with 26 to 75 FSI, and Class C is the lowest rating of 76 to 200 FSI. All classes have a code-specified smoke developed index (SDI) limit of 450.

Class A Fire Rating (FSI 0–25)

Materials in this category offer the highest level of fire protection thanks to their slower flame spread. Class A products are often required in areas where fire risk is highest or where people need extra time to evacuate.

Common Class A Materials

- Gypsum wallboard
- Brick and masonry
- Cement board
- Fire retardant treated wood products
- Certain insulation products, and even paper & plastics

Class A fire-rated materials are often required in egress routes, elevator shafts, stairwells, and exterior walls of multi-family or high-occupancy buildings where there are more people to consider.

Typical locations/sites:

Hospitals, schools, high-rises, egress corridors

Class B Fire Rating (FSI 26–75)

Class B materials provide moderate resistance to flame spread and are typically used in settings with controlled access or moderate traffic.

Common Class B Materials

- Certain types of untreated wood products
- Phenolic foam boards
- Polyurethane insulation with fire retardants
- Treated acoustical ceiling tiles
- Petrol, oil paints, spirit, chemicals etc

These fire-rated materials are often found in offices, hotels, and restaurants where regulations are still strict, but Class A materials aren't mandated.

Typical locations/sites:

Offices, conference centers, mid-rise structures

Class C Fire Rating (FSI 76–200)

Class C fire rating classification is for materials with the fastest flame spread. While not suitable for high-risk zones, they may be allowed in detached structures or areas with lower occupant loads and fire risk.

Common Class C Materials

- Certain types of untreated wood products
- Particleboard
- Certain plastics and composite panels
- Foam panels without protective coatings
- Cooking and Welding Gas
- Electrical fires

Class C-rated materials are often seen in storage sheds, garages, or outbuildings where minimal occupancy and local codes allow their use.

Type of locations/sites:

Storage areas, warehouses, detached garages

Cause for concern:

Failure to meet the International Building Code (IBC) and other local building codes also has significant ramifications, where permitting can be denied or changes to structures must be made to meet compliance.

For example, using a Class C fire-rated material in a required Class A corridor can jeopardize occupancy approval, even if the building is otherwise compliant.

Guidelines:

Fireproof implies zero combustion, which is something very few standard construction material ever achieves.

Flame spread relates to surface burning characteristics, while fire resistance relates to how long a wall, floor, or ceiling assembly can prevent the spread of fire and withstand structural failure.

ASTM E84 lab testing of materials at a site/location, can provide a standardized benchmark that building codes and safety professionals rely on to ensure minimum performance thresholds are met across the board.

The research paper and its concurrent detailing try to address issues found or commonly experienced in different afflicted / incidence prone locations/sites.

The research paper details a need to design SMART RS&A-RCS packages or use EOCC linked

RS&A-RCS services, like identifying or contracting partners in V2R effectiveness with or without earlier lifecycles of flame spread testing and fire compliance

Portable Hand-held analytics/easily mounted device with air quality monitor recommendations can also help evaluate / report High and low temperature conditions, humidity levels, industrial gas level incidences, hazards in underground mining conditions, etc

The UL 1715 is the Standard for Fire Test of Interior Finish Material. It's also known as the room-corner test.

A full-scale room structure is used to evaluate the flammability contribution of wall material assemblies, ceiling material assemblies, or both, exposed to early fire growth under specified room fire exposure conditions.

The room corner fire tests go beyond the flame spread index and smoke developed index provided by Steiner Tunnel testing to ASTM E84 or ANSI/UL 723.

These are examples of projects that could use the UL 1715 test:

- **Commercial:** Office and retail buildings
- **Industrial:** Warehouses and distribution centers
- **Residential:** Multi-family apartments and condos

The UL 1715 fire test works with a range of materials, including wall and ceiling finishes, coatings, and composite systems

Condition evaluation

Thermocouples/gas/air quality sensors are mounted in specific locations in an area or location, to measure the temperature of the plume above the ignition source and other areas.

Photographic and video recording evidence of the fire growth within the area or it's hypothesis based area-coverage and smoke development is important for any pattern documentation.

The use of a visually aided design/model for the area/location structure and the designed area's/location's corner burn fixture can ensure test conditions help expectation or mimic typical building environments

There is no limiting smoke concentration measurement or requirements in UL 1715.

The in-time of incidence/post incidence inspection can evaluate the char in the remains of the burnt material/composite material, to report any diminishing distance for spread from the fire source.

Perspective imagery/ sensing

Consider the perspective sensing/imagery is associated with a closed area ,single entrance or and fixed window scenario, though it may be otherwise

Check for known or existing site/location/case scenario documentation

Evaluate the flame spread and fire contribution to interior finish materials

Report fire risk / hazard result upon first review

Repeat similar settings-based evaluation in five or more minutes

Complete the RSA-RCS reporting

Evaluation Criteria

In the UL 1715 test, the RSA-RCS reporting will need to summarize on fire performance parameters like (1) temperature development within the actual site/location/room or perspective designed one, (2) flame progression, and (3) time to flashover.

The RSA-RCS policy and Methodology tabulations will need to determine the need for fire extinguisher and fire detection & protection products like

1. Stored pressure type extinguishers for Class A, B and C fires

2. Carbon dioxide type extinguishers for Class B fires

3. Foam type extinguishers for Class A and B fires

4. Water type extinguishers for Class A fires

5. Clean Agent fire extinguishers for sensitive electronic equipment Class A, B and C fires

6. Automatic ceiling mounted fire extinguishers for enclosed spaces, unoccupied areas or rooms specific Class A, B and C fires

7. Temporary or permanent water inlet outlet configuration + Short brand pipe with nozzle products like the swinging type wall mounted hose reel drum

8. Site/Location/Area/Room configuration for fire risks/hazards based Automatic sprinklers

9. Supportive water delivery hoses for fire-fighting applications

10. Fire extinguishing or scenario specific service accessories like the fire man axe, the fire man hook, fire bucket with or without stand, fire blanket, categories of nozzles, hose box, mcp push button type service accessories, and mcp break glass with chain service accessories

11. Fire Alarm systems like Heat detectors, Smoke detector Air quality monitors, Humidity level monitors

12. Fire suppression systems for electrical panels, server rooms, computer rooms, document stores, Tape/DATA storage cabinets, telecommunication, switch gear, vaults, process equipment, instrumental occupied or unoccupied electronic installation areas

13. Fire suppression systems for kitchen fires or cooking gas or oil based fires, where wet chemical based extinguishers or suppression systems with post-use ease of cleanups, put out fires in locations, or in ducts, and hoods that are installed

14. During fire incidence or as a pre-requisite based clothing and accessories for occupational health & safety, electrical safety, open-ended or closed loop traffic safety,

15. Emergency Response products like Gas detectors, Emergency First Aid Kit Box, Stretchers, Escape Smoke Heads, Self-contained breathing apparatus, Lifeline AED Automatic Defibrillators, Lifeline Oxygen packs, Nomex suits, Fire Proximity Suits

16. Emergency Exit Lights of designs to suit fire incidence response

17. Fire Evacuation Plans on the Bulletin Board and in locations of fire incidences, where the building or site plan is detailed with Concerning Emergency Response details like Location at this instant, Fire extinguishers available, First Aid Kits available, Emergency Exit direction, Emergency Exits available, **Perspective imagery/ sensing** Monitors, Sirens, Flashing lights available

18. RS&A-RCS policy enabled **Societal Welfare- Emergency Response Phone Numbers** details like phone numbers for the Police, For the Fire & Emergency Services departments, for Ambulances and for Disaster Management departments

19. RS&A-RCS policy for ....

Possible must-do(s)

- ✞ A RS&A package must have a strategy to correlate V2R effectiveness impacting details
- ✞ A RS&A package must have a strategy to design V2R strategies and risk mitigation
- ✞ A RS&A package must have a strategy to conduct random sampling of V2R effectiveness or VSL information
- ✞ A RS&A package must have a strategy to conduct random sampling of V2R performance information like Driver Fitness, Vehicle Fitness, Road System understanding, Alpha Assistance contribution and Call to attension methodology improvement, so there is no issue of misinformation in reporting effectiveness, responsiveness and planned performance.

20. RS&A-RCS policy for

ASTM E84 lab testing essentials

UL 1715 fire test essentials

Decisions and onboarded availability of Portable Hand-held analytics/easily mounted devices that can evaluate / report High and low temperature conditions, humidity levels, gas level incidences, and lift room/shaft/underground or basement structure hazards

Cause for concern guidelines like fireproofing, smoke development and fire contribution to interior finish materials

21. As hypothesis testing,

The RS&A-RCS policy or hypothesis is evaluated for V2R Effectiveness for

Fire identification/prevention/protection, responsiveness and planned performance

The hypothesis testing is done for case scenarios using the V2R Effectiveness equation

$Y = \alpha + \beta i$

where Y is the overall V2R effectiveness - performance predictor for a RS&A-RCS case scenario

α is the Technical Performance Measure related Alpha coefficient, here it is taken as 0.95

βi is the Beta coefficient for RSA-RCS policy and Methodology for V2R/Cause for concern effectiveness,

responsiveness but not Veritably-packaged for Risk Mitigating Intelligent or SMART Methodology

Function information like Driver Fitness, Vehicle Fitness, Road System understanding and

Alpha Assistance participation & any unplanned delay in agility

The surveys at the briefest level pose Top … insightful questions such as

1. The RS&A-RCS policy must have a strategy to design, and decide on Methodology tabulations for V2R strategies (this is in context with understanding, participation and minimum unplanned reactions)

Response: Yes/No/Partially Yes

2. The RS&A-RCS policy must have a strategy to design, and decide on Methodology tabulations for V2R risk mitigation (this is in context with fire detection & protection products/solutions)

Response: Yes/No/Partially Yes

3. The RS&A-RCS policy must have a strategy to design, and decide on Methodology tabulations for correlating impact to V2R effectiveness (this is in context with(Fire prevention & safety) FP&S performance measures for concerns like unknown or unregulated fire sources, fire development, fire spread or progression and no reviewed earlier "fire evacuation plan, fire detection, suppression & protection products")

Response: Yes/No/Partially Yes

4. The RS&A-RCS policy must have a strategy to call fire & emergency response departments and enable (Fire prevention & safety) FP&S measures for concerns like unknown or unregulated fire sources, fire development, fire spread or progression and no reviewed earlier “fire evacuation plan, fire detection, suppression & protection products”)

Response: Yes/No/Partially Yes

5. The RS&A-RCS policy must have a strategy to call close-by or linked assisting fire & emergency response groups/centres and enable (Fire prevention & safety) FP&S measures for concerns like unknown or unregulated fire sources, fire development, fire spread or progression and no reviewed earlier “fire evacuation plan, fire detection, suppression & protection products”)

Response: Yes/No/Partially Yes

6. The RS&A-RCS policy must have a strategy to decide upon and install First Aid Kits, Emergency Exit availability and directions, and (in-sync with trends) “**Perspective imagery/ sensing** Monitors, Sirens, and Flashing lights to warn and help evacuate people”

Response: Yes/No/Partially Yes

7. The RS&A-RCS policy must have a strategy to decide upon and use “fire test essentials for tabulated hazardous conditions” to determine and enable and enable (Fire prevention & safety) FP&S measures

Response: Yes/No/Partially Yes

<u>Tabulated hazardous conditions</u>

1. Unregulated fire sources or storerooms or go-downs

2. Unregulated fire spread or progression

3. No valid fire evacuation plan

4. No system to sound an alarm or siren for a fire incidence/fire development

5. No norm or system to know the type of emergency response fires, that is Class A, or B or C fires